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Please Welcome,

Kumar Nachiketa, APJ Storage SA - WWPS

What's New with Amazon FSx for Windows File Server?

13th August 2020

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What's New with Amazon FSx for Windows File Server?

Learn the latest capabilities of Amazon FSx

Kumar Nachiketa, APJ Storage SA, WWPS

Date 13th August 2020

I just want to access my files….

I just want to access my files….

But I end up managing it all

Agenda

- What is Amazon FSx for Windows File Server?
  - Overview of the service – benefit, features, availability
  - Recently released features
- See Amazon FSx for Windows File Server in action
  - Demo
- Wrap Up
  - Call to action
  - Resource
  - Q&A

Poll #1

Q1# Your background on File storage/ Amazon FSx for the windows file server. Which statement describes the best?

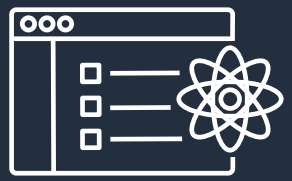
A. Using Amazon FSx for the windows file server, have experience in on-premises NAS

B. New to Amazon FSx for the windows file server have experience in on-premises NAS

C. Never worked on file storage

Amazon FSx for Windows File Server overview

What is Amazon FSx for Windows File Server?



Fully managed native Windows file systems

Deeply integrated with AWS

Fully managed means you no longer need to…

Manage hardware

Plan capacity

Procure and purchase hardware

Set up storage servers and volumes

Detect and address hardware failures

Invest CapEx

Manage software

Install and configure server software

Apply Windows updates

Manage software licenses

Manage backups

Monitor security

Designed for wide spectrum of use cases

NAS lift-and-shift

Home directories

Line-of-business applications

Web serving and content management

Software development environments

Backup and disaster recovery

HA SQL Server databases

Designed for wide spectrum of use cases

Cloud-native workloads

Media workflows

Analytics

SaaS applications

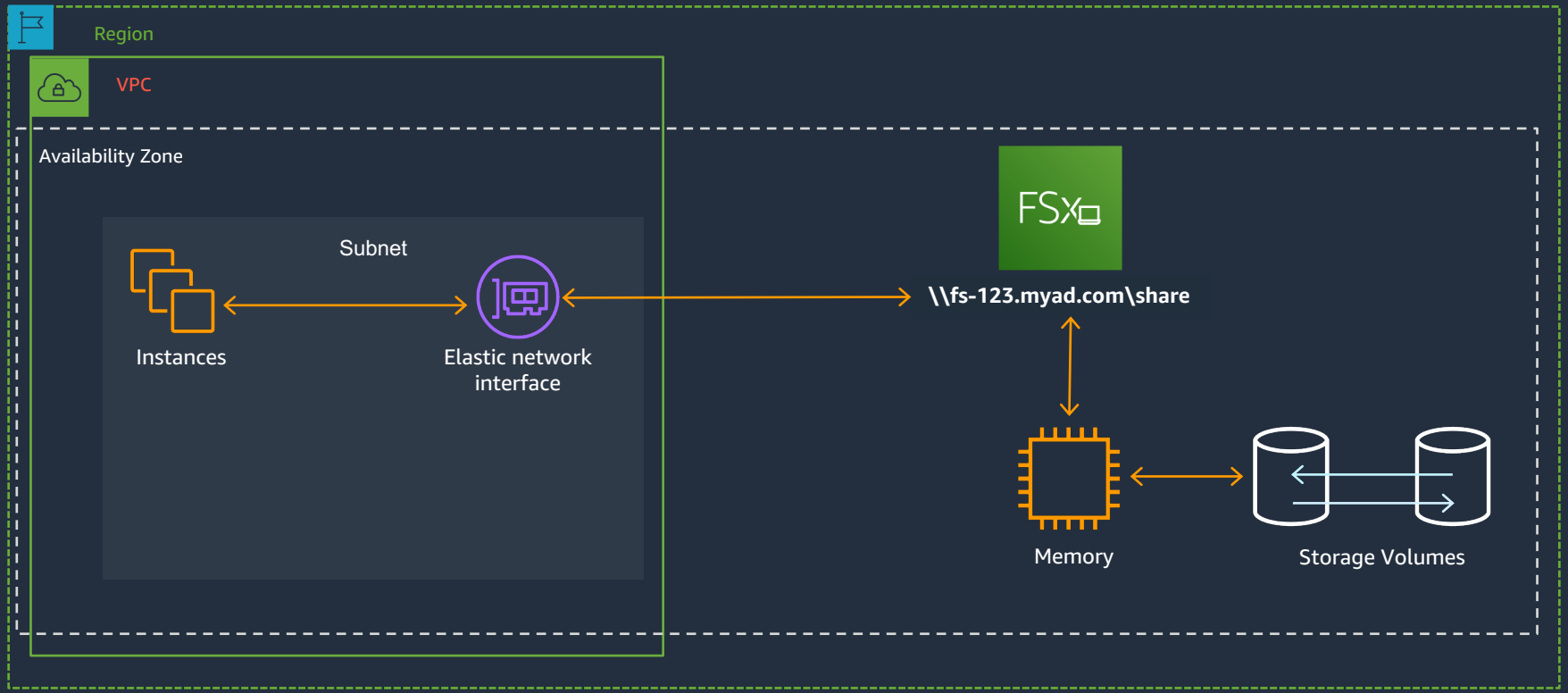
What fully-managed, fully-featured, and compatible means for customers

*"With Amazon FSx we have a **managed solution** that eliminates the need for us to manage the tasks associated with operating file storage, including performing backups and applying software patches."* -Greg Bell, Senior Systems Administrator

*"We no longer need to worry about maintenance of the file system environment or patching of the underlying servers. This is allowing the team to **focus on more business-critical parts of our application structure**."* -John Waller, Senior Engineer, DocXellent

Architecture

Amazon FSx for Windows File Server Architecture



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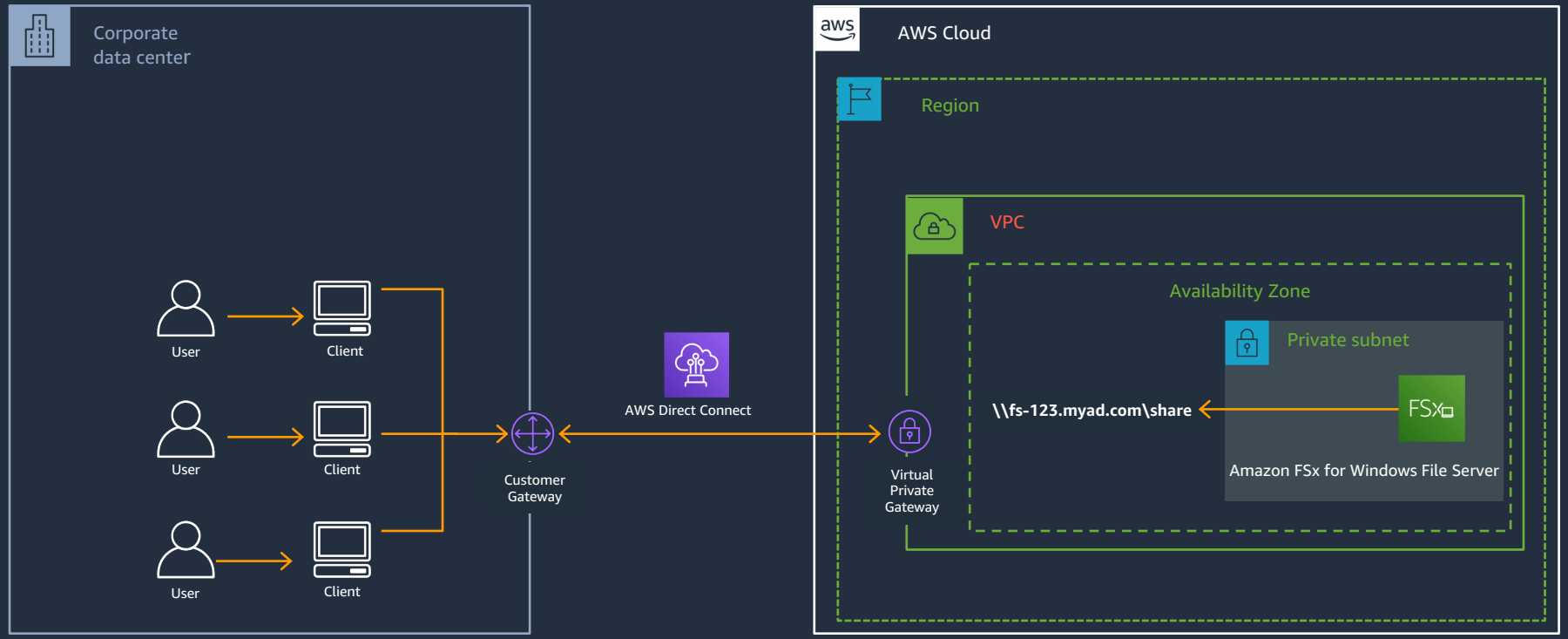
Home directory use case example: Hybrid Architecture



File system deployment options

Single-AZ

Continually monitors and addresses hardware failures

Replicates data within Availability Zone

Multi-AZ

Continually monitors and addresses hardware failures

Replicates data within Availability Zone

Replicates data across Availability Zones

Automatically fails over across Availability Zones

Poll #2

Q2# You can only access FSx for windows file server share within AWS VPC?

A. True

B. False

Poll #3

Q3# Do your customers often use SQL Standard license (vs Enterprise)?

A. YES

B. NO

Dive deep on these aspects

1. Performance & Scale
2. Cost Effectiveness
3. Backup
4. Security
5. Administration

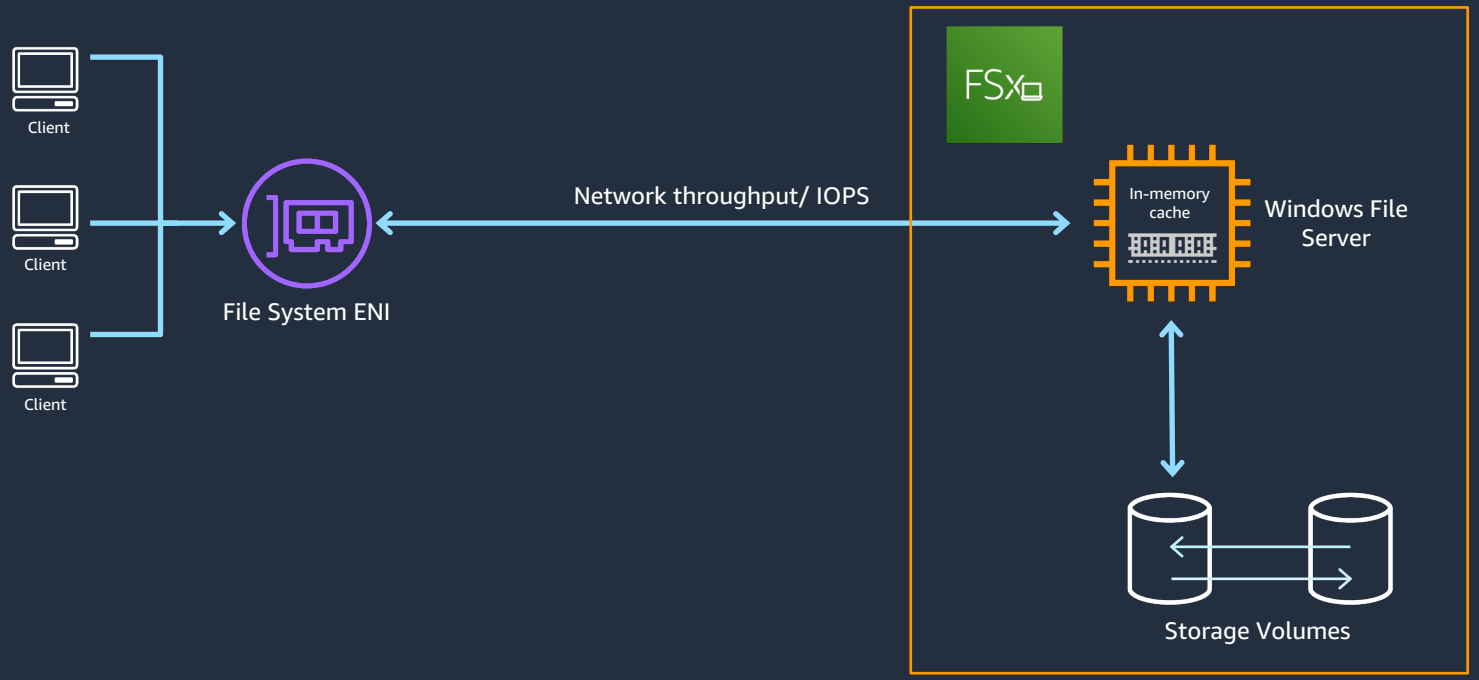
1. Performance & Scale

< 1ms latencies

Multiple GB/s

Hundreds of thousands of IOPS

Architectural components of Amazon FSx file system



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Storage capacity consideration

Storage Type	Disk Throughput (MB/s per TiB of storage)	Disk IOPS (IOPS per TiB of storage)
SSD	750	3000
HDD	12 baseline; 80 burst (up to a max. of 1 GB/s per file system)	12 baseline; 80 burst

Throughput capacity, Baseline and Burst level

FSx throughput capacity (MBps)	Network throughput capacity (MBps)		Network IOPS	Memory (GB) – for caching	Disk throughput (MBps)		Disk IOPS
	Baseline	Burst			Baseline	Burst	
8	8	600	Thousands baseline; tens of thousands burst	0.5	8	260	Hundreds to thousands baseline; tens of thousands burst
16	16	600		1	16	260	
32	32	600		2	32	260	
64	64	600	Tens of thousands baseline	4	64	350	
128	150	1,250		8	128	600	
256	300	1,250	Hundreds of thousands baseline	16	256	600	Tens of thousands baseline
512	600	1,250		32	512	–	
1,024	1,500	–		64	1,024	–	
2,048	3,125	–		128	2,048	–	

Throughput capacity, Baseline and Burst level

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256	300	1,250	Hundreds of thousands baseline	16	256	600	Tens of thousands baseline
512	600	1,250		32	512	–	
1,024	1,500	–		64	1,024	–	
2,048	3,125	–		128	2,048	–	

2. Cost Effectiveness

Storage Pricing

(per GB-month)

SSD

HDD

	Single-AZ	Multi-AZ
SSD-based storage	15.6 cents	27.6 cents
HDD-based storage	1.6 cents	3.0 cents

Note: Based on Amazon FSx for Windows File Server pricing in Singapore region

Data Deduplication

Large datasets often have a lot of duplication, which increases storage costs

User shares (home directories)
Multiple users have many copies or versions of a file

Software dev shares
Most portions of binaries remain unchanged from build to build

Use Data Deduplication to reduce costs associated with duplicated data

Scenario	Content	Typical space savings
User documents	Office documents, photos, music, and videos	30-50%
Software dev shares	Software binaries, build files, and program symbols	70-80%
General file shares	Mix of the above	50-60%

Effective storage cost with data deduplication

(per GB-month with 50% storage deduplication savings)

	Single-AZ	Multi-AZ
SSD-based storage	15.6 cents	27.6 cents
(with deduplication)	7.8 cents	13.8 cents
HDD-based storage	1.6 cents	3.0 cents
(with deduplication)	8.0 cents	1.5 cents

SSD

HDD

Note: Based on Amazon FSx for Windows File Server pricing in Singapore region

Typical savings from deduplication for general file shares is 50-60%

Data Deduplication – How it works?

Works at the sub-file level

Uses post-processing optimization to minimize performance impact

Removes duplicated content and compresses common content

Use remote management PowerShell CLI on your file system to…

- Enable/disable Data Deduplication
- Customize schedule for deduplication jobs
- Monitor how much savings you're achieving with deduplication

```
PS C:\Windows\system32> Invoke-Command -ComputerName amznfsx[redacted].fsx-ad.com -ConfigurationName FsxRemoteAdmin -ScriptBlock {
>> Get-FSxDedupStatus } | select OptimizedFilesCount,OptimizedFilesSize,SavedSpace,OptimizedFilesSavingsRate

OptimizedFilesCount OptimizedFilesSize SavedSpace OptimizedFilesSavingsRate
------------------- ------------------ ---------- -------------------------
                779         1666284377  929007343                        55
```

Example TCO

Storage requirements

- 10 TB of storage
- With deduplication, 50% of storage needed
- Deployment type: Multi-AZ
- Storage type: HDD

Throughput requirements

- 16 MB/s sustained, 100 MB/s burst

Backup requirements

- Expected backup storage usage: 1x of storage capacity

File system component	Total cost
Storage (Multi-AZ, HDD, 5TB @ $0.025/GB-mo)	$128
Throughput capacity (128 MB/s @ $4.50/MBps-mo)	$576
Total cost (excl. backups)	$200/ month (or $0.02/ GB-mo)
Backups (5 TB @ $0.05/GB-mo)	$256
Total cost (incl. backup)	$456/ month (or $0.04/GB-mo)

Poll #4

Q4# You can get high performance and sub-millisecond latencies for actively accessed data irrespective of storage type (SSD or HDD)

A. True

B. False

Poll #5

Q5# FSx for windows file server provides inline De-duplication

A. True

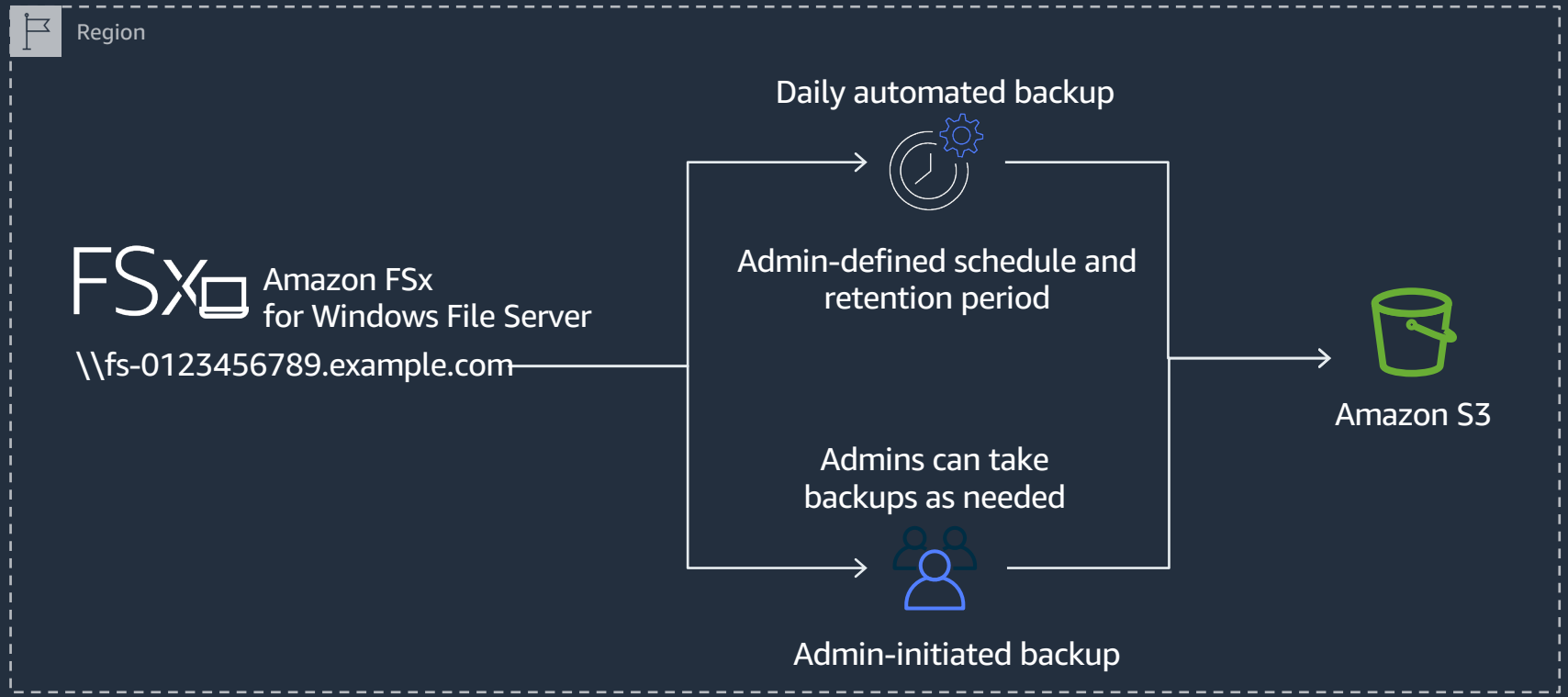
B. False

3. Backup

Shadow copies – file and folder recovery

- FS Snapshot – Point in time copy
- Stored within the file system
- FSx CLI for remote management on
- PowerShell
- On-demand and scheduled
- 512 shadow copies any given time
- Set max size in bytes or percentage
- Individual files or folder

Backups


Region
Daily automated backup
Amazon FSx
for Windows File Server
\\fs-0123456789.example.com
Admin-defined schedule and
retention period
Amazon S3
Admins can take
backups as needed
Admin-initiated backup

Backups

Highly durable	Highly durable (11 nines) – stored in Amazon S3
File system consistent	Capture and restore a point-in-time view of file system Ensures file system-consistency using VSS
Incremental	Only changes after your most recent backup use backup storage
Fully managed	Automatic daily backups, with retention policy Admin-initiated backups via API/Console

Shadow copies and backups compared

Feature	Shadow Copies	Backups
Storage Location	Within the file system	Amazon S3
Storage Capacity	Configurable size (% or Bytes)	Unlimited
Retention	Up to 512 at a given date	Automatic – upto 35 days User Initiated - unlimited
File System Consistent	YES	YES
On-demand	YES	YES
Scheduled	YES	Automatic – Once in a day User Initiated – customized schedule
User-Initiated Restore	YES	NO
File and folder restores	YES	NO
File System restores	YES – File Shares	YES – New File System

4. Security

Security and compliances

Data encrypted
at-rest and in-transit

Option to enforce
encryption in-transit

Integrates with
your organization's
AD and supports
Windows ACLs

Network traffic access
control using Amazon VPC
security groups

Admin API
access control
using AWS IAM

Monitor and log
API calls using
AWS CloudTrail

PCI-DSS + ISO-
+ SOC + GDPR compliant
and HIPAA eligible

Use FSx with your organization's AD

Directly integrate your FSx file systems with your organization's active directory (on-premises or in-cloud)

Authentication: Your users continue to access file shares by authenticating with their existing AD user credentials

Authorization: You can migrate and use your existing file and folder ACLs, and your share-level access control as is, without any modifications needed

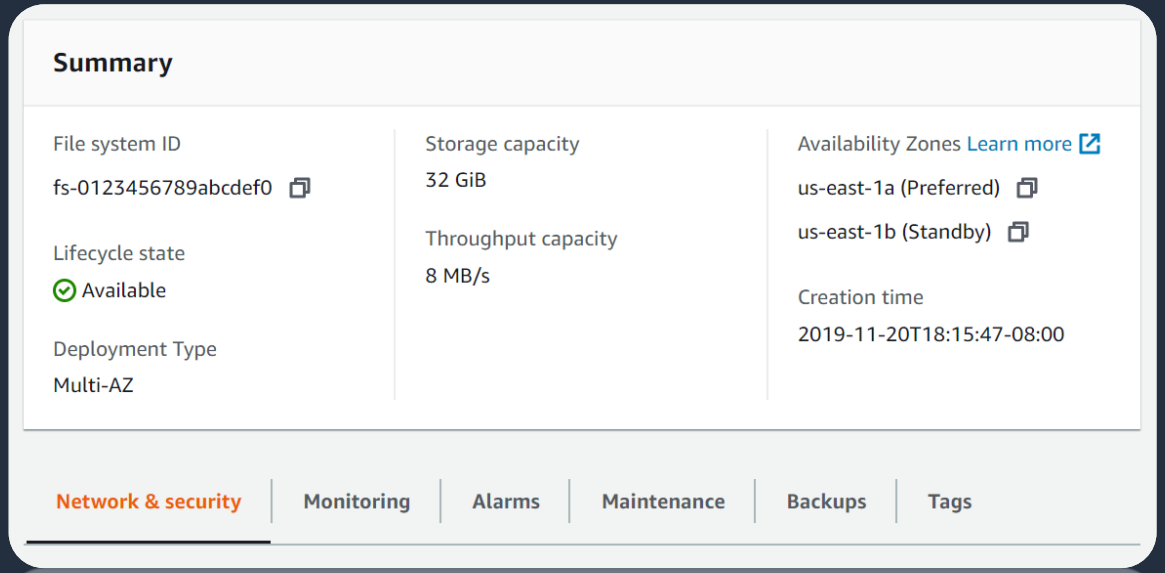
Supports two AD integration options:

- AWS Managed Microsoft AD
- Self-managed Microsoft AD (on-premises or in-cloud)

5. Administration

Administering AWS resource properties

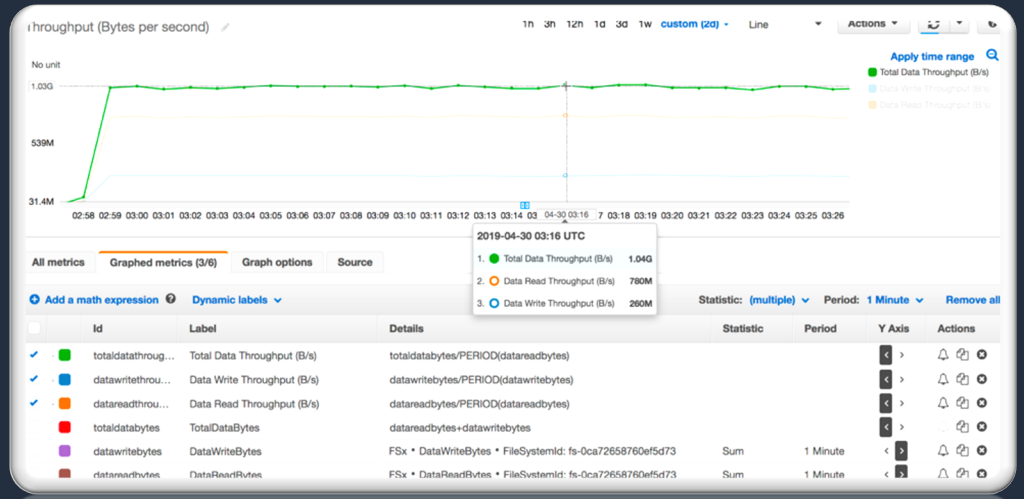
- CloudWatch metrics
- CloudWatch alarms
- Maintenance window
- Automatic backups
- Tags
- Active Directory configuration



CloudWatch metrics

Every minute, Amazon FSx emits metrics to Amazon CloudWatch:

- Free storage capacity
- File system throughput (read & write)
- File system operations (read, write, and metadata)



You can use Metric Math to simplify near real-time monitoring of your file system

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Administering file system using remote Powershell

What you can configure using remote PowerShell:

- SMB file shares
- Shadow copies
- Data deduplication
- User quotas
- Open sessions / files
- Enforcement of encryption in transit

Migration

What is AWS DataSync

Simplifies, automates, and accelerates your online data transfer

Migrate active application data

Transfer data for timely processing or Archiving

Replicate for data protection and recovery

Transfers up to 10 Gbps per agent

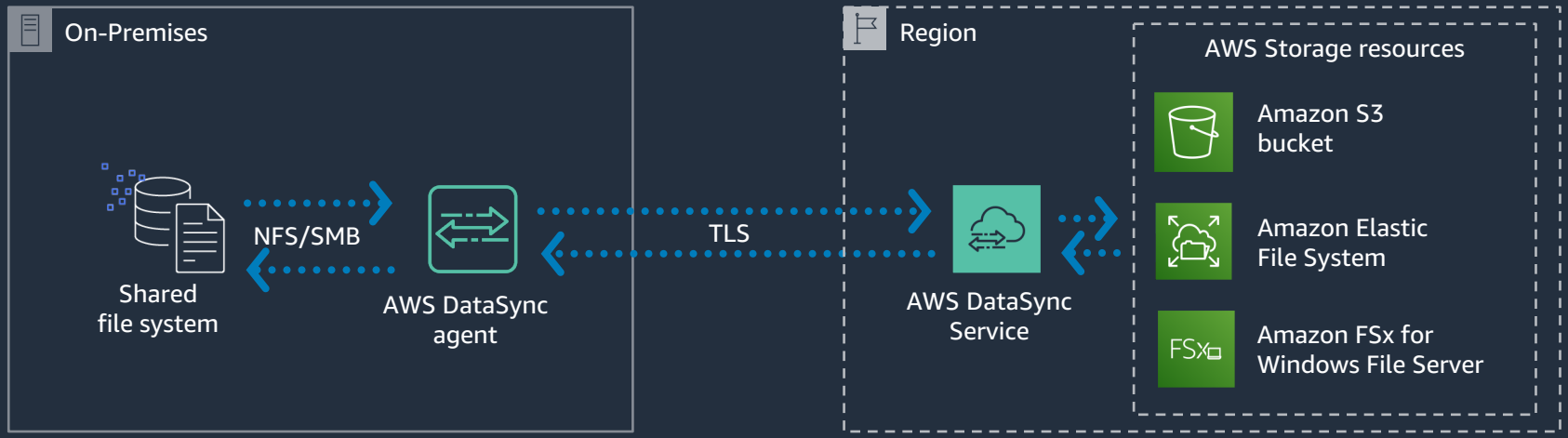
Simple data movement to S3, EFS, FSx for Windows

Secure and reliable transfers

AWS integrated

Pay as you go

How AWS DataSync Works?



Highly parallel transfers using optimized network protocol

Fully-managed service scales to send/receive data from agent

Optimized read & write to S3 & EFS using VPC Endpoints

The speed and reliability of *network acceleration* software with the cost-effectiveness of *open source tools*

DEMO

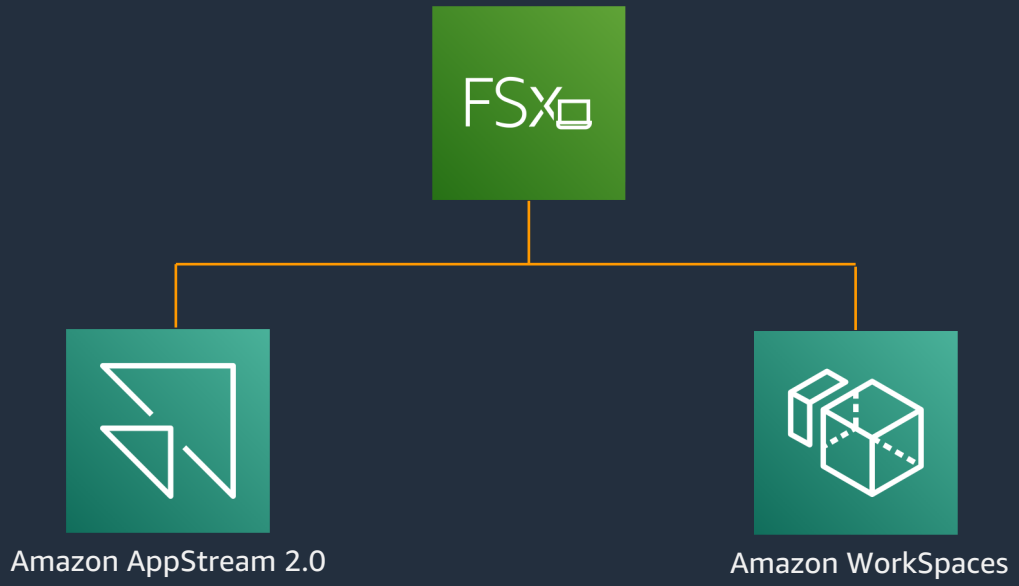
Poll #6

Q6# Do you have an End User Computing consulting offer on AWS (e.g. Workspaces, AppStream 2.0)

A. YES

B. NO

End user computing use case with FSx for Windows File Server



Blogs
AppStream 2.0 https://aws.amazon.com/blogs/desktop-and-application-streaming/using-amazon-fsx-with-amazon-appstream-2-0/
Workspaces https://aws.amazon.com/blogs/desktop-and-application-streaming/using-amazon-fsx-for-windows-file-server-with-amazon-workspaces/

Next steps for migrating file shares to AWS

1. Discuss your architecture and migration path with an AWS Solutions Architect

2. Setup a test using our deployment guide: https://aws.amazon.com/blogs/storage/using-amazon-fsx-for-windows-file-server-with-an-on-premises-active-directory/

3. Watch our re:Invent deep dive: https://www.youtube.com/watch?v=_x_Geur93oc

4. Watch our webinar: Reduce File Storage TCO with Amazon EFS and FSx: https://www.youtube.com/watch?v=jx2OqfAoalQ

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Thank You!

Please join us again for another PartnerCast session

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